IS216 G5 · Team 8

SG Handyman Booking

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Kong Jia Zhen Amanda

Video	*https://youtu.be/bBZYex9hspA?si=-onliRcwlN42shFK*
Live app	https://sg-handyman.vercel.app






SANITON
SANITON

Google

help fixing toilet singapore

SparkFlow Pte Ltd
https://www.sparkflow.com.sg › Services › Plumbing

Toilet Repair Singapore

We diagnose the cistern, fixture and drain, then quote repair or replacement. **Call +65 8163 5626** … a fix Quick Quote 9010 Tampines Street 93, #02-125 Singapore ...

Homejourney
https://www.homejourney.sg › Blog › Home Services

Ultimate Toilet Repair Guide: Common Problems & Fixes

29 Dec 2025 — The Homejourney guide addresses **common toilet repair issues in Singapore**, such as flushing failures, leaks, and clogs, pricing starting at S$50 ...

Discussions and forums

Poor plumbing service experience in Singapore with unclogging and toilet ...

Facebook · Complaint Singapore · 40+ comments · 2 years ago

This shouldn't be a problem if you wash instead of wipe.
More ›

Top answer · 10 votes · 2 yea...

You all eating too much fiber? 🤣 More ›

2 votes · 2 years ago

Should not have used the first service provider you come across. It is better to call a ...
More ›

2 years ago

What are affordable options for toilet bowl replacement or repair in ...

Facebook · Seedly Personal Finance Commun... · 9 comments · 3 years ago

Is there a good way to get a stronger toilet flush in HDB?

Reddit · r/singapore · 40+ comments · 5 years ago

See more ›

88 24 Hours Plumbing SG

Hello, I need urgent help. My toilet flush in my HDB isn't working since morning. Tried doing it myself on Google but to no avail. 9:21 PM

When is the earliest time u can come over n help me take a look? 9:22 PM

Pls help, this is very urgent 9:22 PM

My address is 10A Toa Payoh #12-183 9:22 PM



Hello, thank you for reaching out to 24 Hours Plumbing SG. My name is Gladys and I'm happy to serve you today!

Unfortunately our earliest slot will be 10th December. We are fully booked out for this week.

Can I book you for a slot next Thurs?

01
The Problem

Who struggles, and why

The problem

Most new homeowners have no trusted contractor. Handyman and plumbing firms each run their own website or rely on word of mouth, so no single place lets owners compare them. Bookings happen over WhatsApp or phone, so an owner must contact firms one by one to find a free slot and a price, while slots in a busy estate fill up fast. Prices are not published in a standard way, and reviews on company websites can be filtered or inflated.

Some jobs also need a licensed firm: since 2018, it is an offence to hire a plumber who is not licensed by PUB for regulated plumbing work, but owners cannot easily check who is licensed.

Pain Point #1:
No one place to search for firms

Pain Point #2:
Can’t view available & upcoming slots

Pain Point #3:
Unverified reviews

Pain Point #4:
Difficult to find PUB-licensed firms

Pain Point #5:
Prices not clear and upfront

02
Our Solution

An aggregated handyman booking site

SG Handyman Booking

A one-stop platform that brings handyman and plumbing firms into a single place. Homeowners can compare starting prices across companies, read reviews from verified customers only, and see each firm's live availability, so there's no need to message firms one by one on WhatsApp.

When they find a suitable timeslot, they can book it directly, and the system makes sure no other customer can take the same slot at the same time.

Not every problem needs a contractor, so the platform also supports DIY repair, with a forum where users share fixes and a shop for common repair tools.

Feature #1:
Company listings

Feature #2:
Verified service reviews

Feature #3:
Live appointment booking

Feature #4:
DIY Forum

Feature #5:
DIY Repair Tool Shop

Tech stack

FRONTEND

Vue 3 + Vue Router
Bootstrap 5
Axios
Leaflet

Builds the pages and talks to our backend

BACKEND

Node.js
Express

Runs our API and business logic

DATA

MongoDB
OneMap API

MongoDB stores our data; OneMap maps office locations

TESTING

Playwright (end-to-end)
Postman (API)

Playwright tests user journeys; Postman tests the API

DEPLOYMENT

Vercel

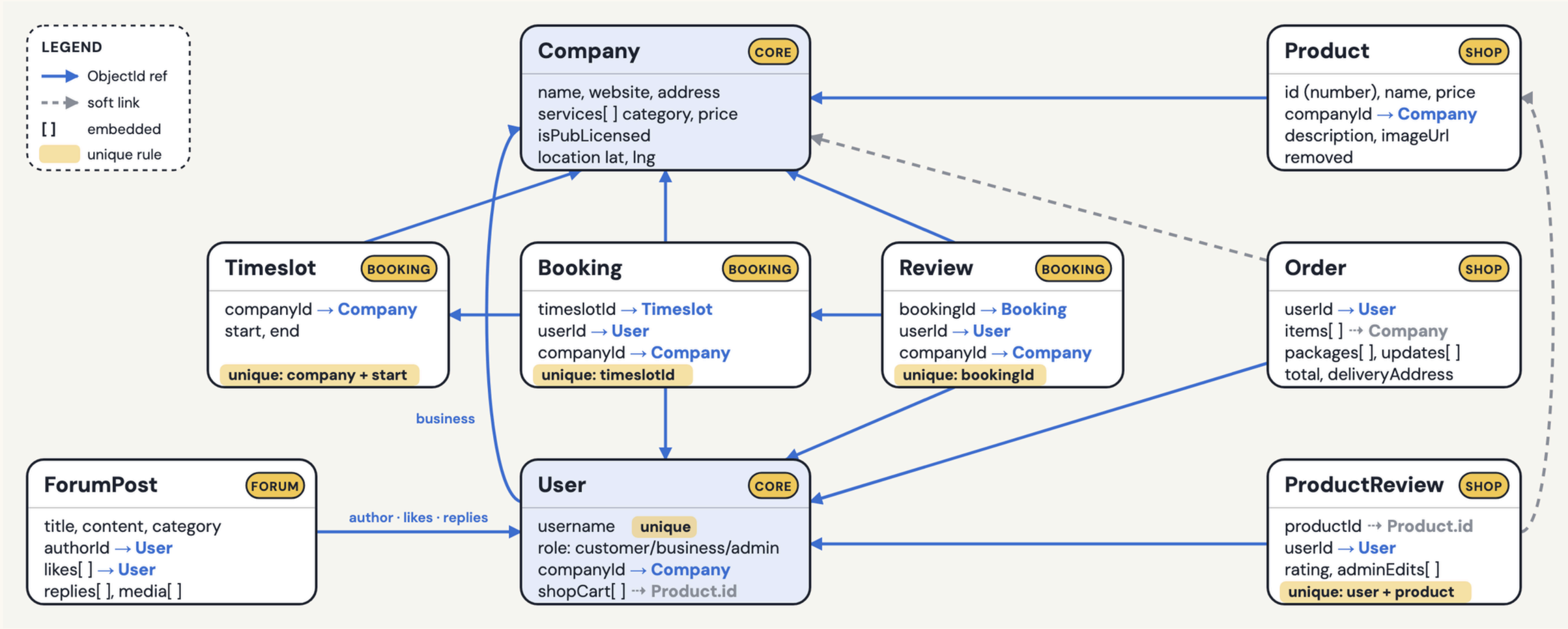
Hosts the live site

COLLABORATION

GitHub

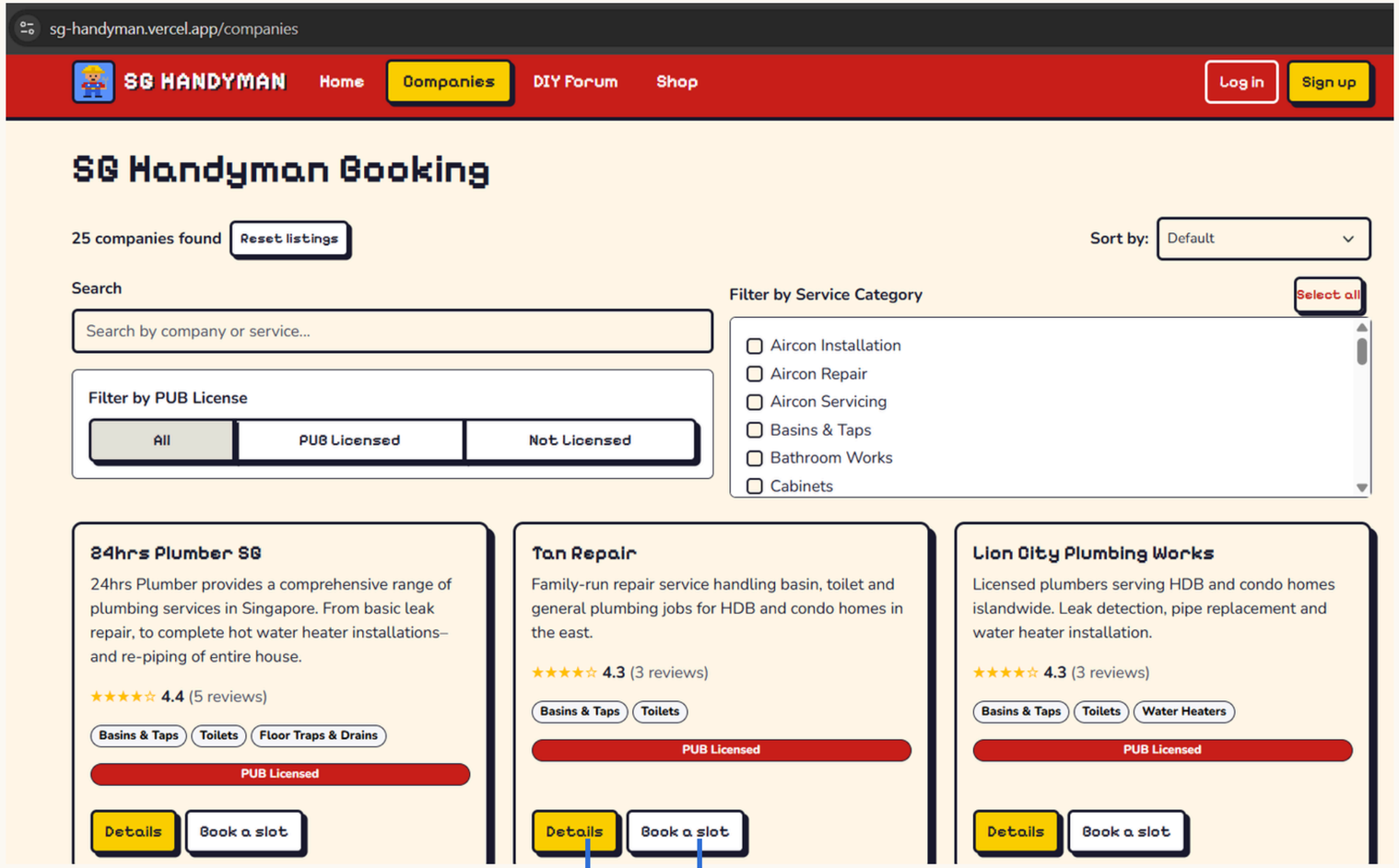
Shared code and version control

Data store


LEGEND
ObjectId ref
soft link
[ ] embedded
unique rule
Company
CORE
name, website, address
services[ ] category, price
isPubLicensed
location lat, lng
Product
SHOP
id (number), name, price
companyId → Company
description, imageUrl
removed
Timeslot
BOOKING
companyId → Company
start, end
unique: company + start
Booking
BOOKING
timeslotId → Timeslot
userId → User
companyId → Company
unique: timeslotId
Review
BOOKING
bookingId → Booking
userId → User
companyId → Company
unique: bookingId
Order
SHOP
userId → User
items[ ] ⇢ Company
packages[ ], updates[ ]
total, deliveryAddress
business
ForumPost
FORUM
title, content, category
authorId → User
likes[ ] → User
replies[ ], media[ ]
author · likes · replies
User
CORE
username unique
role: customer/business/admin
companyId → Company
shopCart[ ] ⇢ Product.id
ProductReview
SHOP
productId ⇢ Product.id
userId → User
rating, adminEdits[ ]
unique: user + product

03
Company Listings

Company Listings



Leads to company details page

Features:

1) Search bar

- Searches by company name and service category

2) PUB License toggle

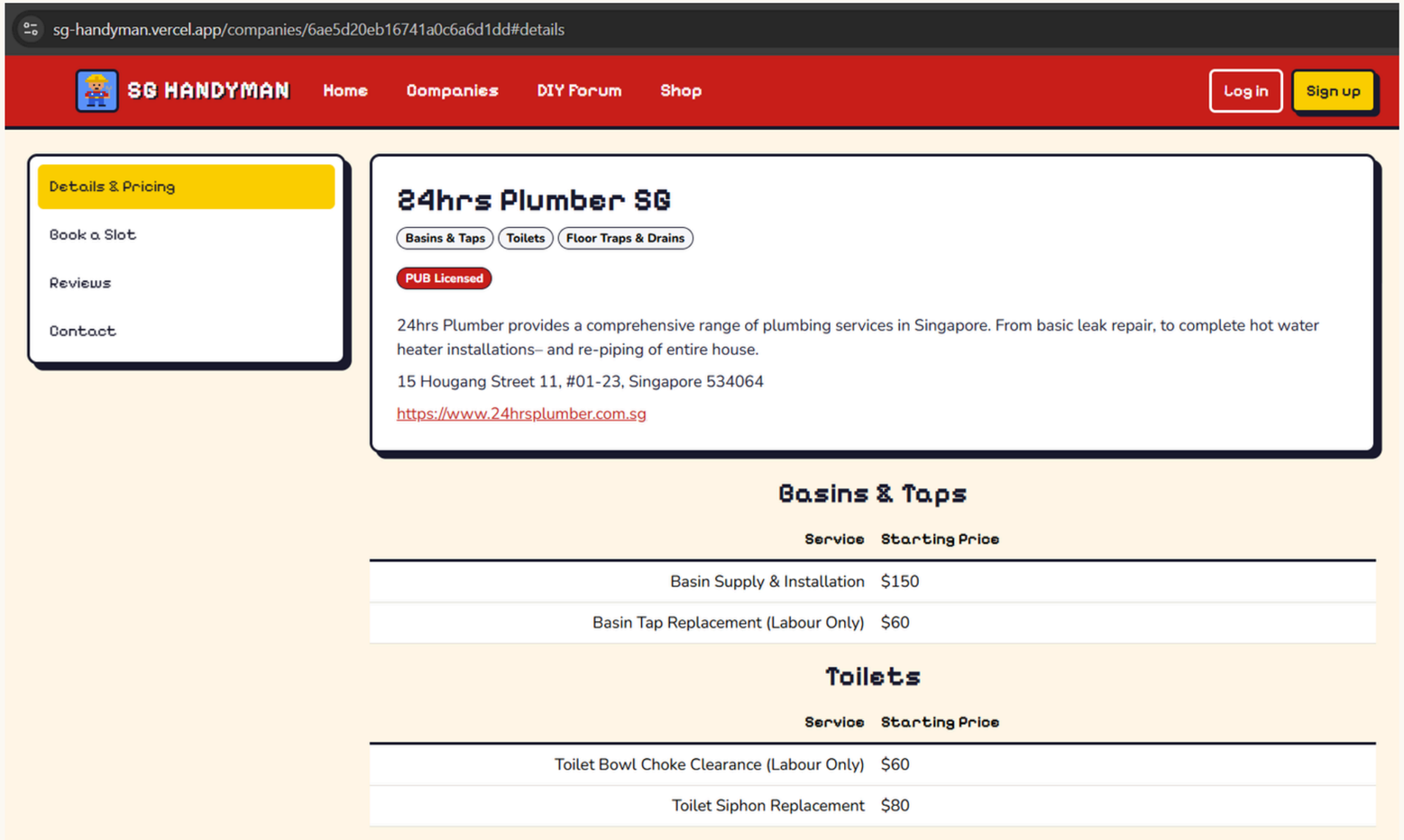
- Filters companies that have at least one PUB licensed plumber

3) Service Category Filter

4) Sort dropdown

- Default
- Earliest free timeslot
- Highest rating
- Latest user review
- Alphabetical Company name

Company Details



Notable Features:

1) Side navbar

- Lets user jump to different sections without scrolling

2) Tables of services offered with starting price

Data store

companies collection


```
_id: ObjectId('6ae5d20eb16741a0c6a6d1dd')
name: "24hrs Plumber SG"
description: "24hrs Plumber provides a comprehensive range
website: "https://www.24hrsplumber.com.sg"
address: "15 Hougang Street 11, #01-23, Singapore 534064"
▾ services: Array (3)
  ▾ 0 : Object (3)
      category: "Basins & Taps"
    ▾ details: Array (2)
      ▾ 0 : Object (3)
          service: "Basin Supply & Installation"
          startingPrice: 150
          _id: ObjectId('6ae5d20eb16741a0c6a6d1df')
        ▸ 1 : Object (3)
      _id: ObjectId('6ae5d20eb16741a0c6a6d1de')
  ▸ 1 : Object (3)
  ▸ 2 : Object (3)
isPubLicensed: true
__v: 0
▾ location: Object (2)
    lat: 1.354712153486001
    lng: 103.8798789412471
```


Notable attributes:

1) address

- Postal code is used to pinpoint office location in OneMap

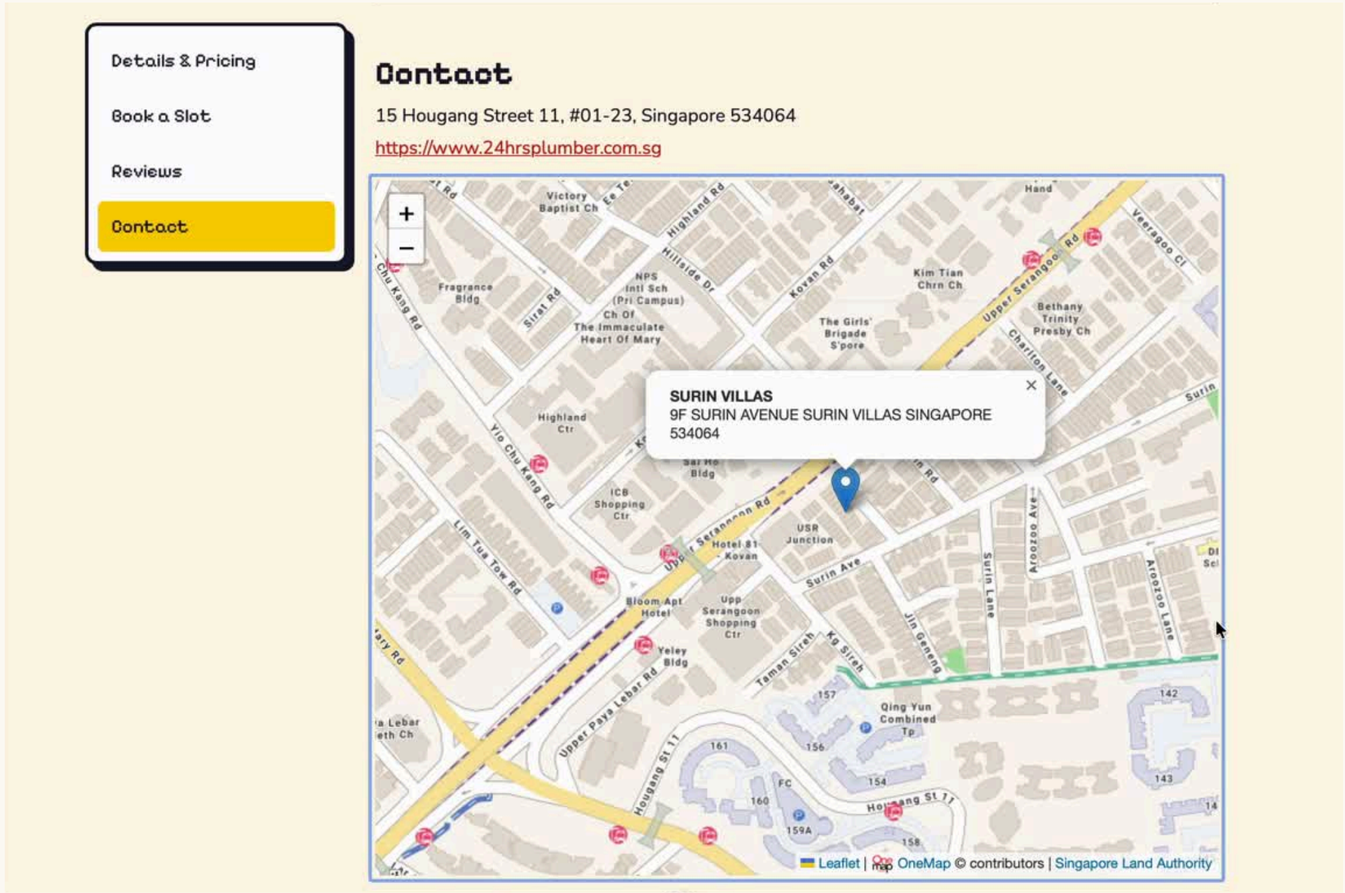
2) name & services > category

- Used for searching and filtering

3) isPubLicensed

- Used for filtering

External API: OneMap (implemented with Leaflet)



How it's implemented

1) Client uses Leaflet and OneMap's api to render the map (does not require OneMap api key)

2) Client sends company's office postal code to server

3) Server queries OneMap's api for map coordinates corresponding to postal code (requires OneMap api key)

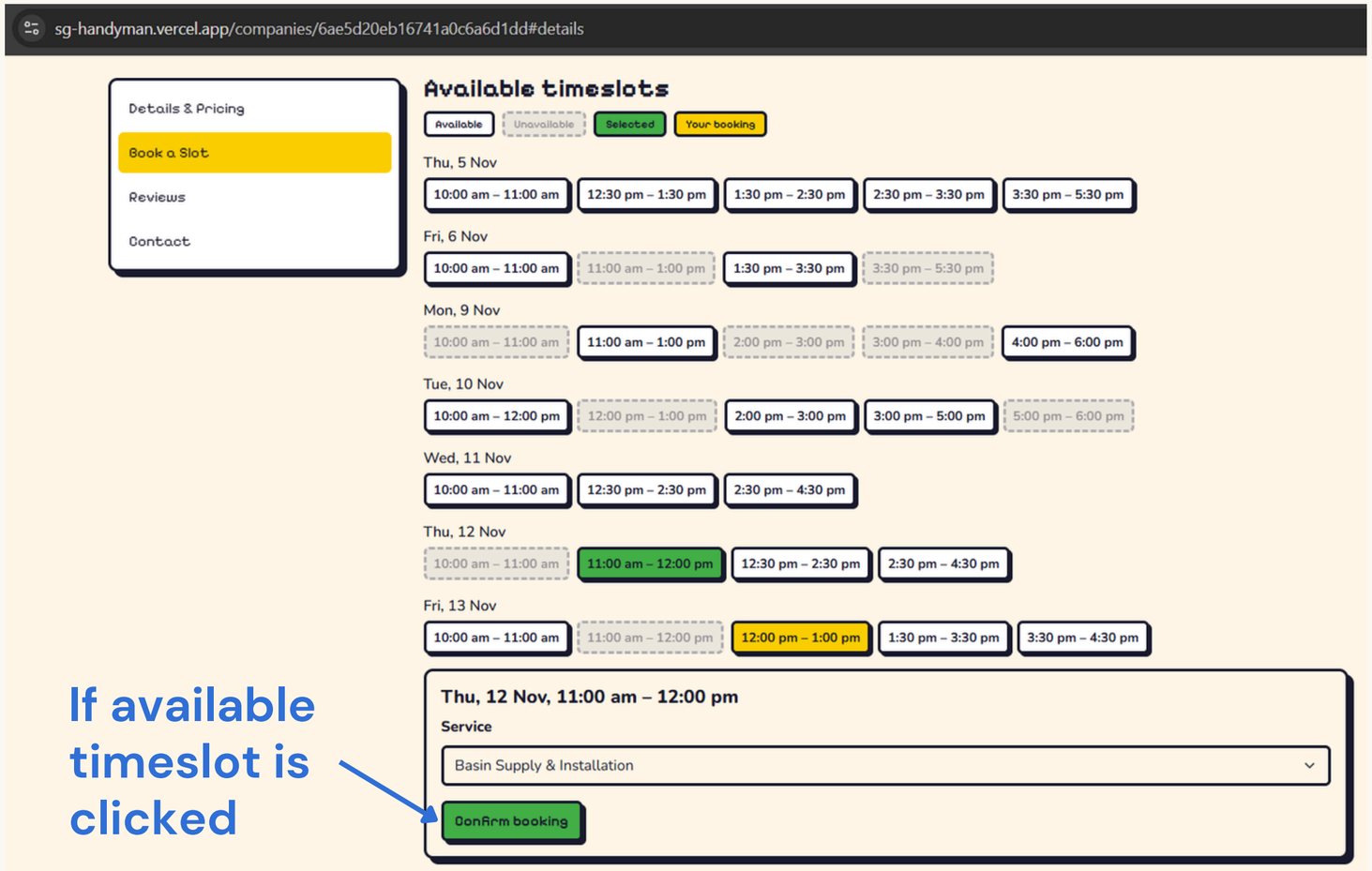
4) Server responds to client with map coordinates

5) Client renders map coordinates

OneMap API key stored in server .env and not in client

04
Appointment Booking

Timeslots



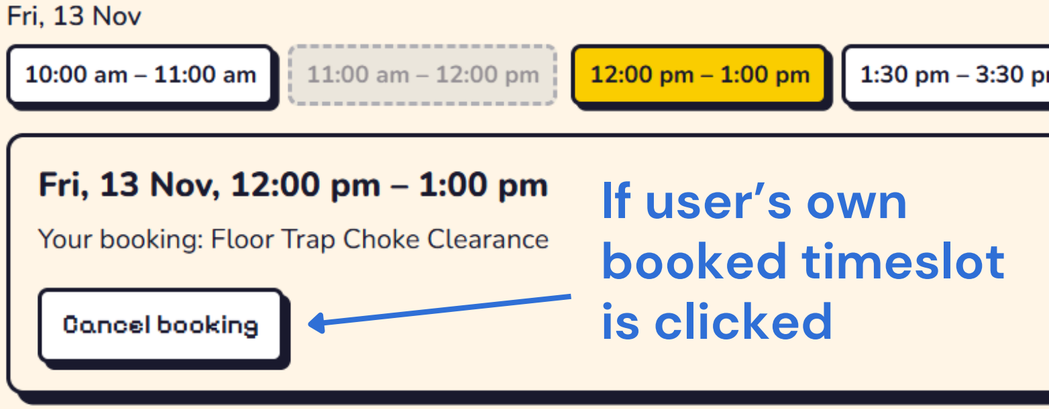
If available timeslot is clicked

Notable Features

1) Timeslots are reactive, will refresh every second to show which timeslots are taken by other customers

2) Server checks if a timeslot is taken before it saves a booking. If taken, reject the booking with a clear message and does not save it (no double booking)

3) Users can cancel their bookings



If user's own booked timeslot is clicked

Data store

timeslots collection

```
_id: ObjectId('6ae5d213b16741a0c6a6d313')
companyId: ObjectId('6ae5d20eb16741a0c6a6d1dd')
start: ISODate('2026-11-02T02:00:00.000+00:00')
end: ISODate('2026-11-02T03:00:00.000+00:00') N
__v: 0
```

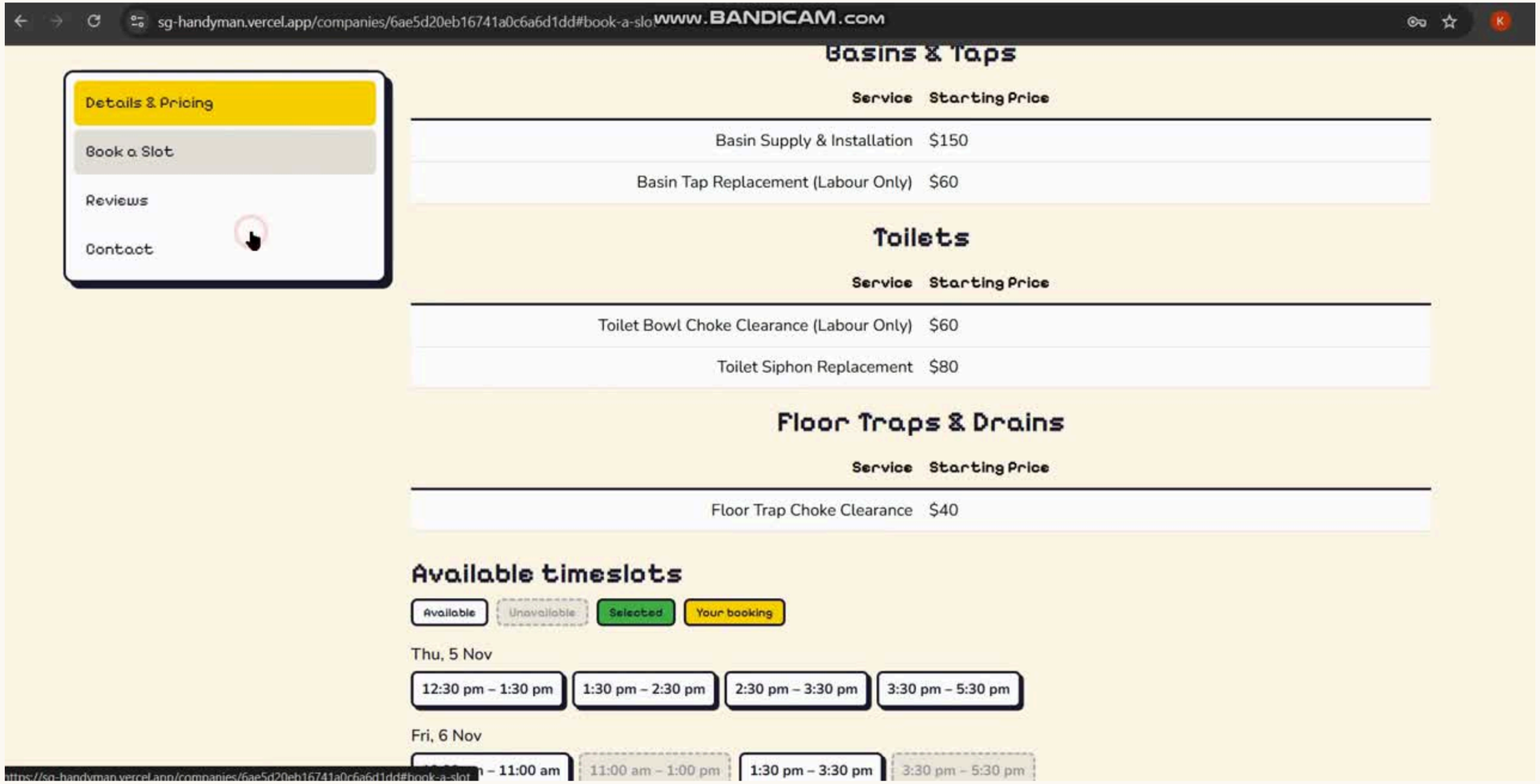
bookings collection

```
_id: ObjectId('6ae5d216b16741a0c6a6d6ea')
timeslotId: ObjectId('6ae5d213b16741a0c6a6d313')
companyId: ObjectId('6ae5d20eb16741a0c6a6d1dd')
userId: ObjectId('6ae5d212b16741a0c6a6d2c0')
service: "Floor Trap Choke Clearance"
createdAt: ISODate('2026-10-31T11:33:42.634+00:00')
__v: 0
```

Timeslots are the available timings that users can book.

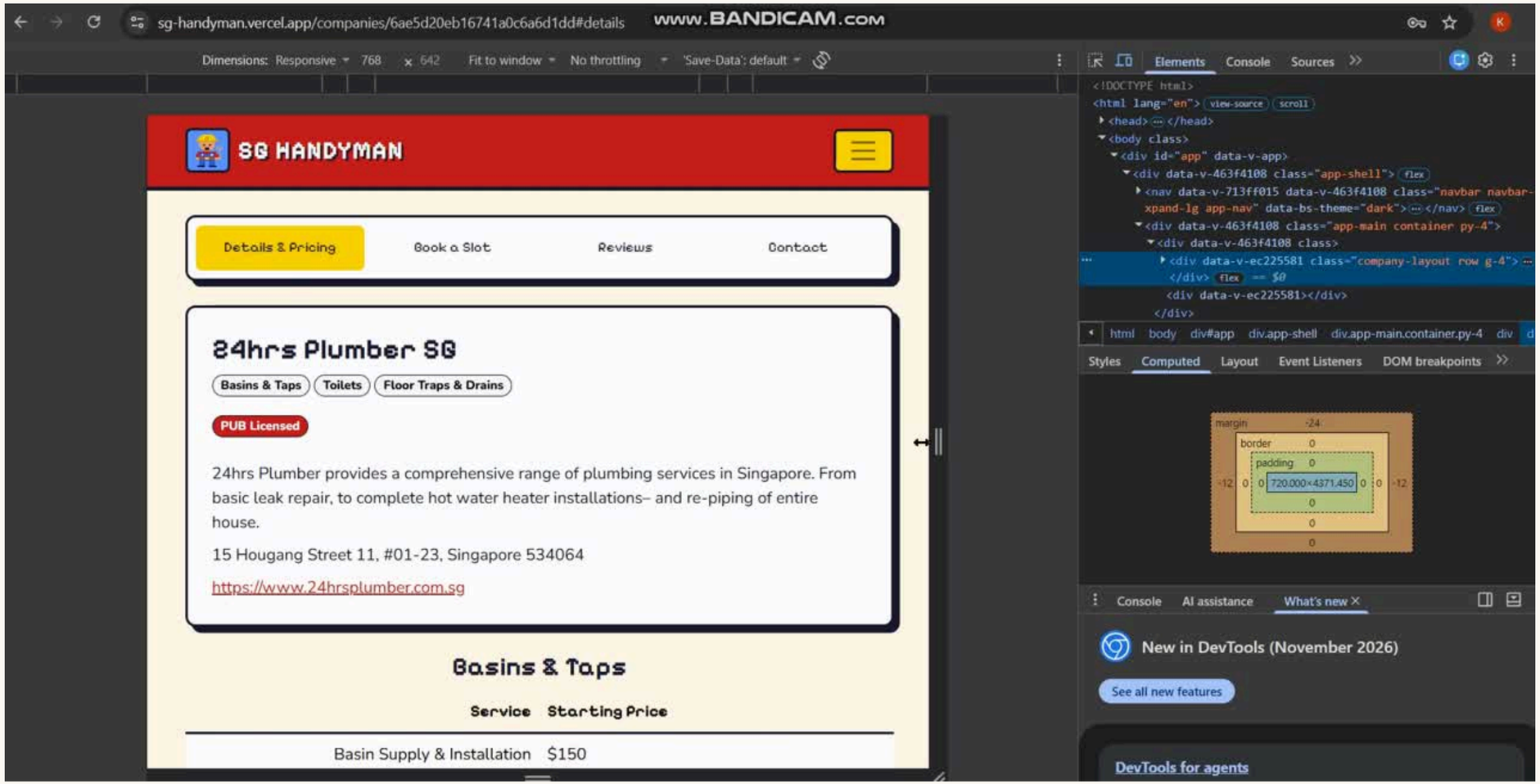
Bookings are the timings that users have booked.

Styling and responsiveness



Side navbar

Styling and responsiveness



Screen width responsiveness

05

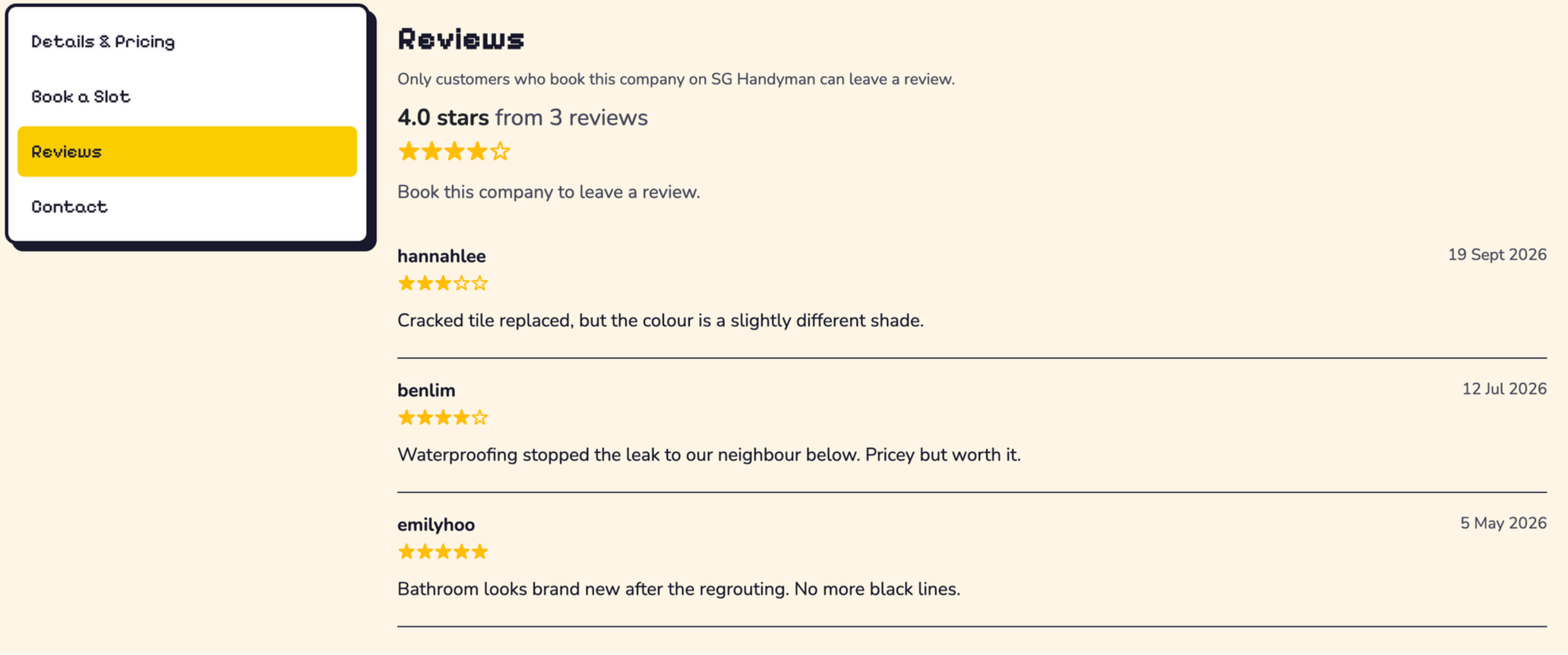
Verified Reviews & My Bookings Page

Verified Reviews

1. Viewing reviews and ratings
2. Leaving a review for a booking
3. Editing and deleting your review
4. Rating on company cards

Verified Reviews

View reviews on a company's page

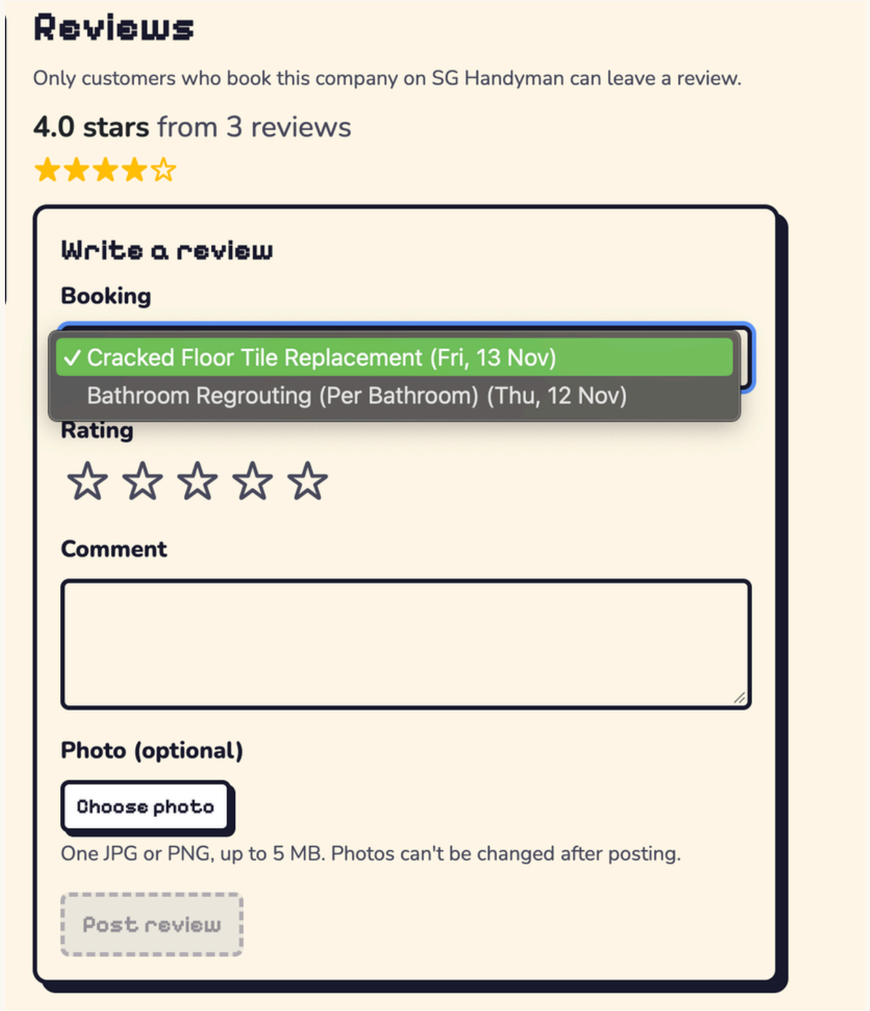


Verified Reviews

Leave a review



Before making a booking at the company



After making a booking at the company

Verified Reviews

Leave a review

Reviews

Only customers who book this company on SG Handyman can leave a review.

4.0 stars from 3 reviews

★★★★☆

Write a review

Booking

Bathroom Regrouting (Per Bathroom) (Thu, 12 Nov)

Rating

★★★☆☆

Comment

staff was late, but job was well done

Photo (optional)

bathroom.jpeg Remove

One JPG or PNG, up to 5 MB. Photos can't be changed after posting.

Post review

Choose which booking to leave a review for, only 1 review per booking allowed

Optionally choose to upload an image to substantiate the review

test-customer-1

Service: Bathroom Regrouting (Per Bathroom)

★★★☆☆

staff was late, but job was well done

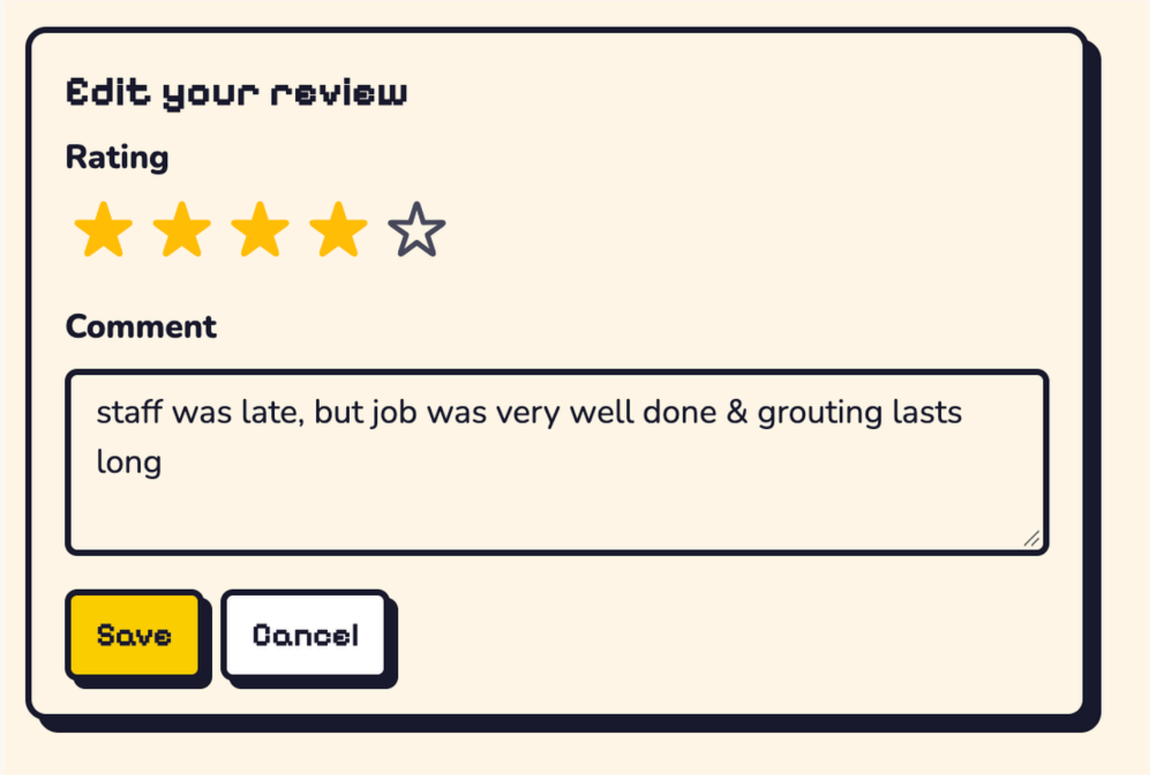


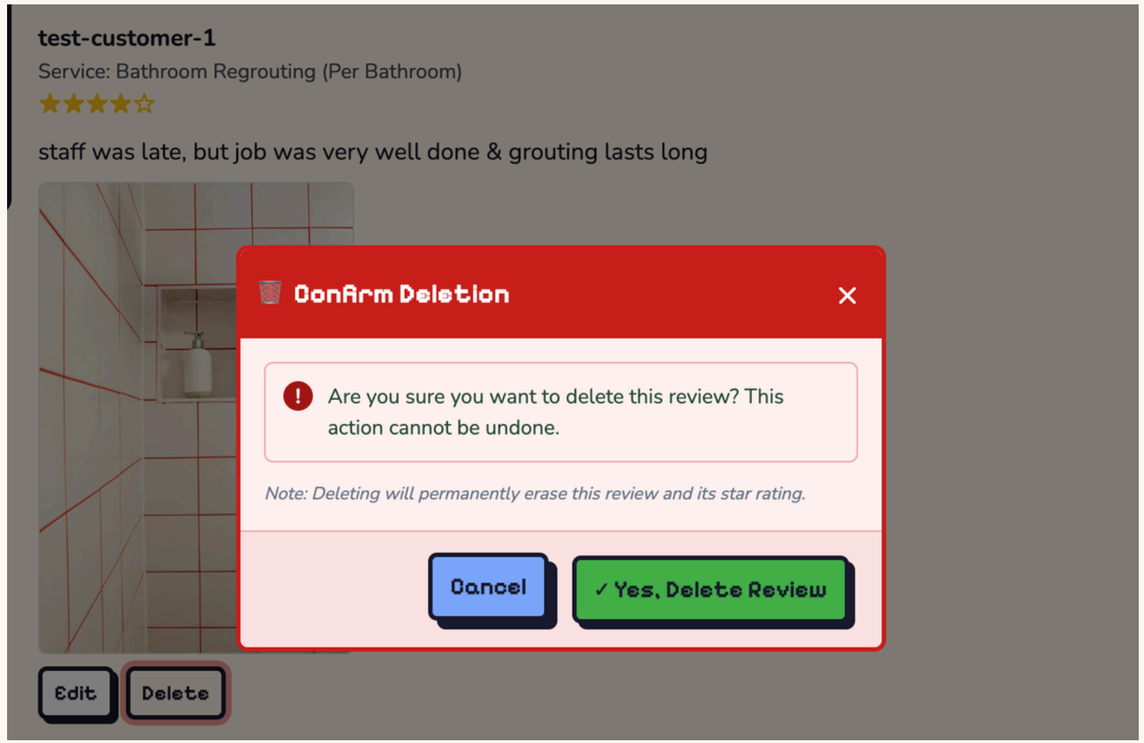
Edit Delete

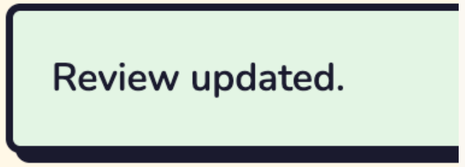
Review appears after being posted

Verified Reviews

Edit & delete review





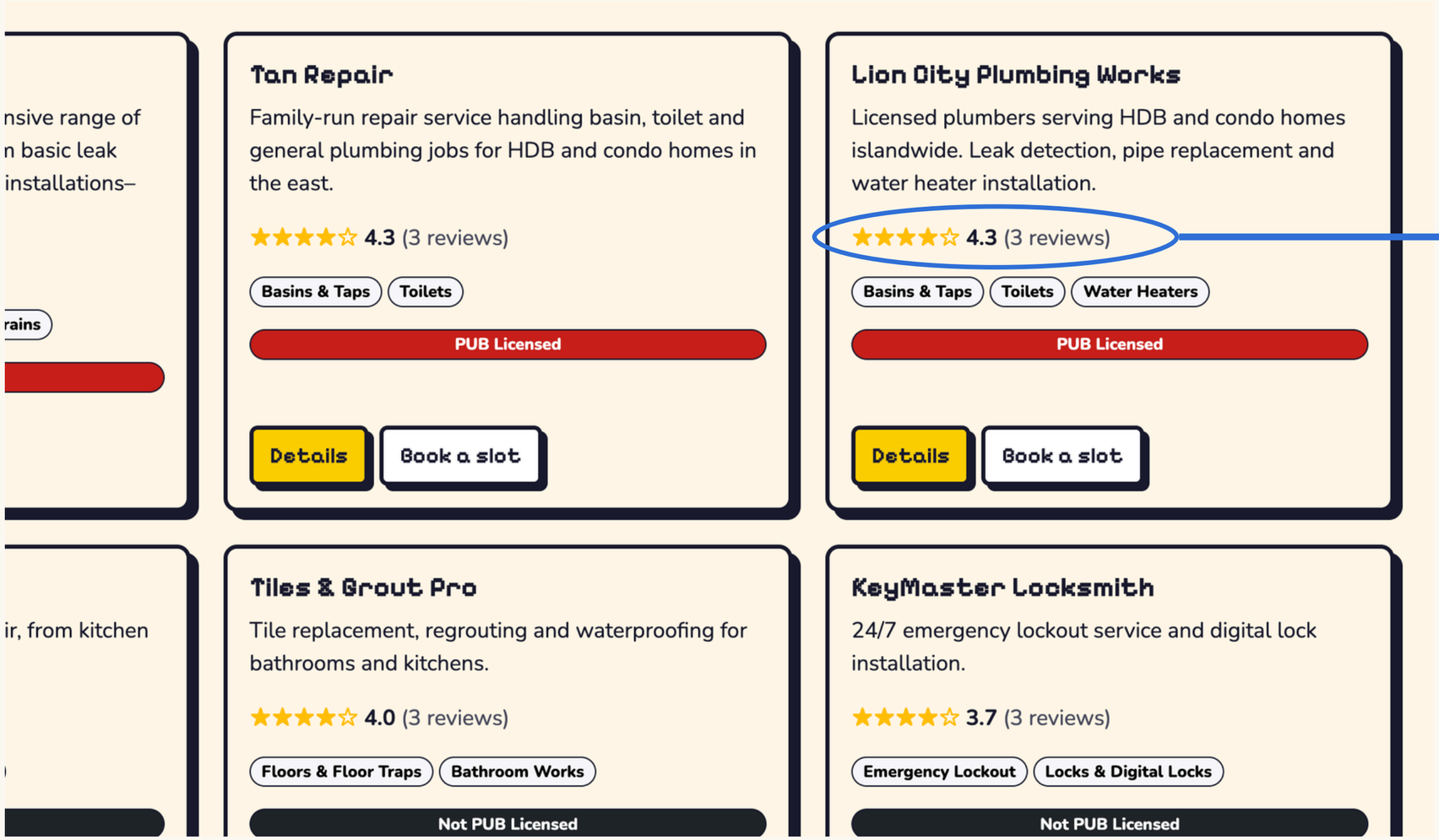


Edit confirmation pops up after ‘save’ is clicked

Delete confirmation pops up when ‘delete’ is clicked

Verified Reviews

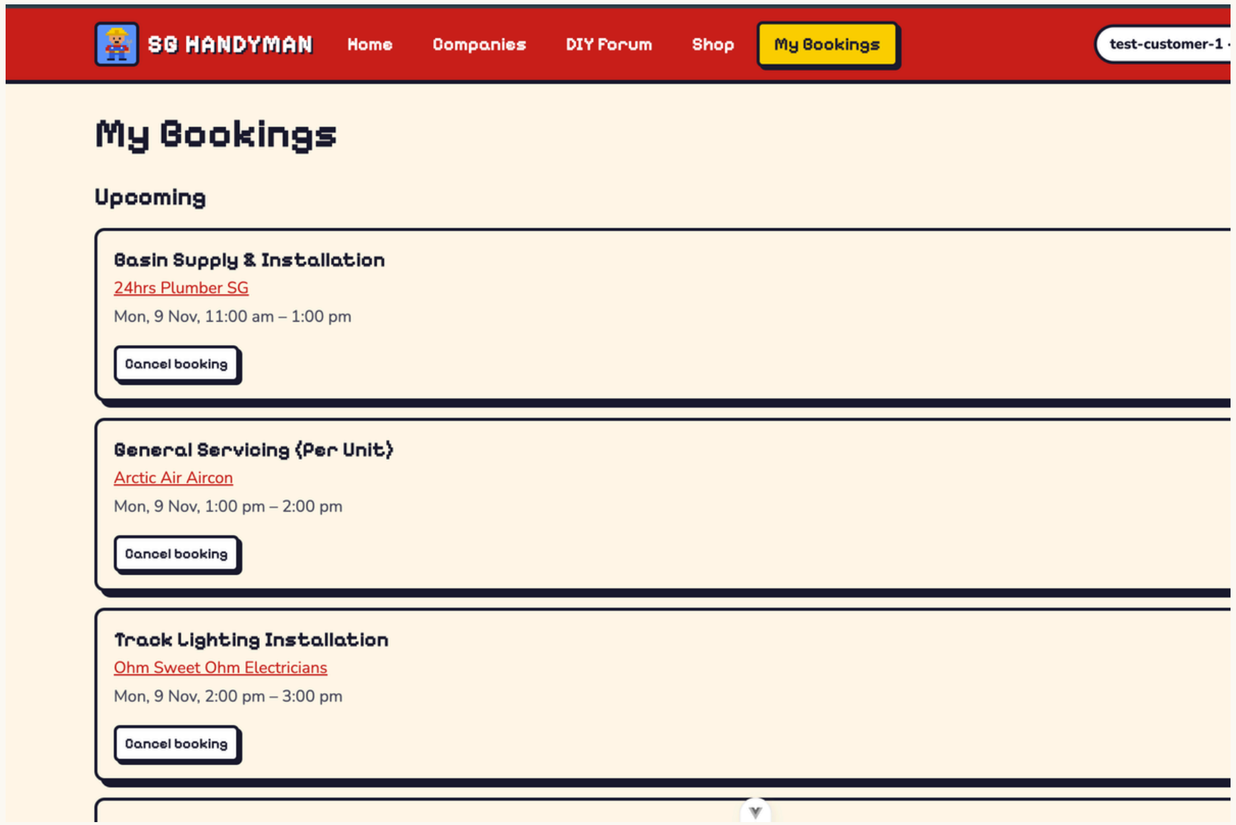
Rating on company cards



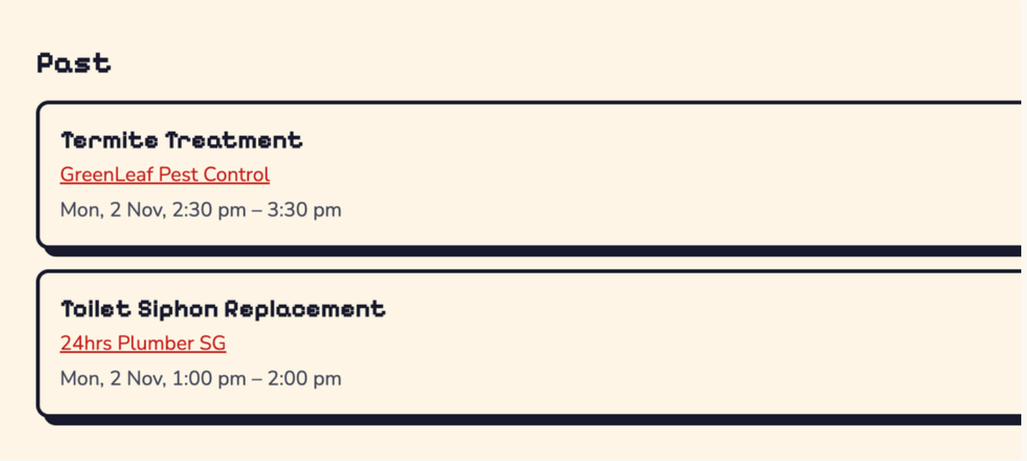
Average rating & number of reviews shown on each company's details card

→ offers viewers comparison in a quck glance

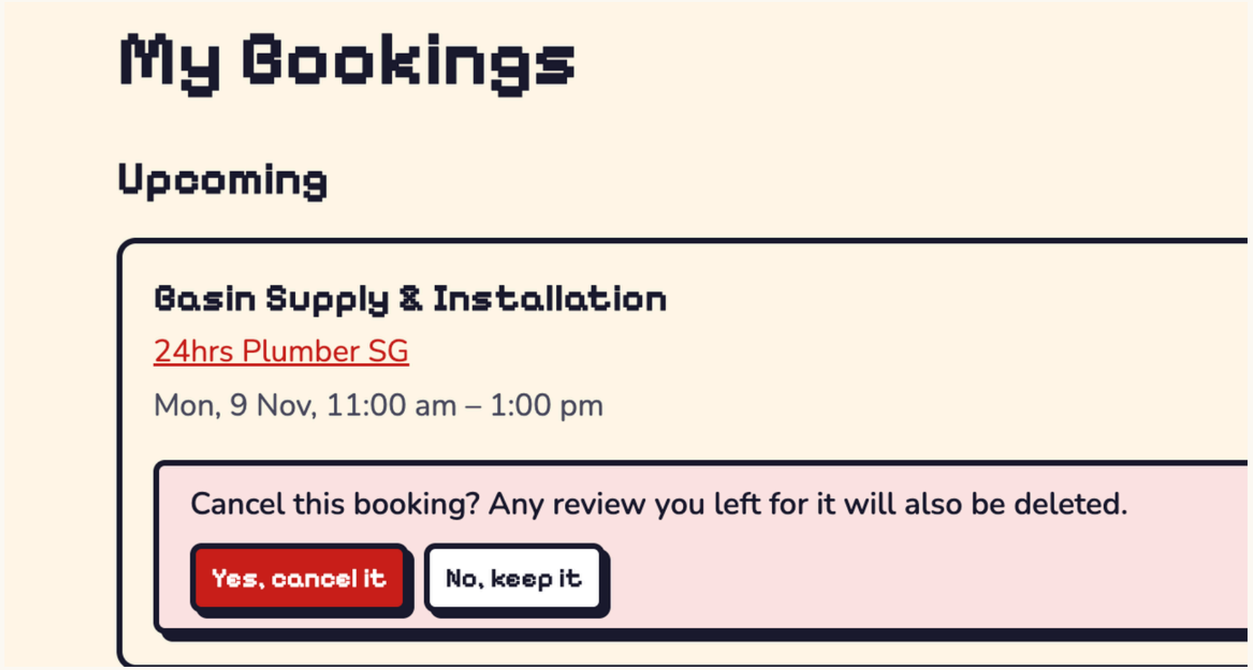
My Bookings



View upcoming & past bookings

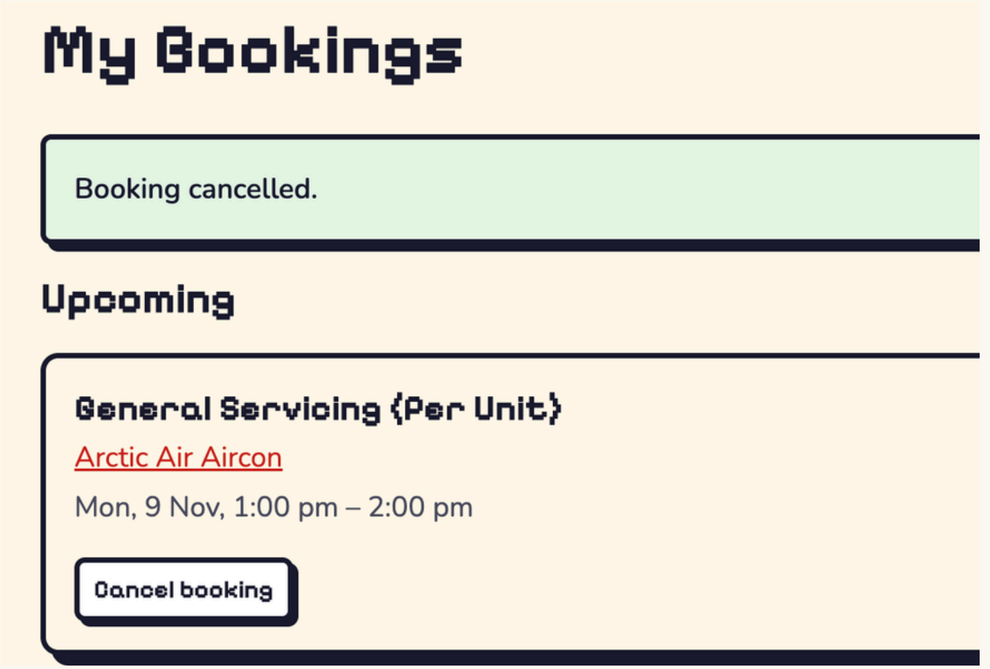


My Bookings



Cancel confirmation pops up when ‘cancel’ is clicked

Cancel bookings



Cancel confirmation appears

06

DIY Community Forum

DIY Community Forum

1. Creating a post / reply
2. Viewing all the posts and replies
3. Updating each post and reply
4. Deleting user’s review and reply

DIY Community Forum

Adding a post or reply

Create a post

Title

What do you need help with?

Category

Leaking tap | Aircon servicing | Power trip | Door lock | Wall mounting | Others

Details

Describe the problem or share your DIY tip.

Add images (optional)

Choose File

Choose one JPG or PNG image, up to 5 MB. Images cannot be changed after posting.

Create post | Cancel

Aircon servicing

Aircon Leaking Water

Should I seek help for this or can I fix this manually by myself?



Posted by test-customer-1 on 11/2/2026, 9:38:12 PM

Was this helpful? 🔥 (1)

Edit post | Delete post

Replies (1) Show replies

Write a reply

Add reply

DIY Community Forum

Viewing posts

Search and Filter Panel for Posts

Posts

Search posts...

Search

Popular: Leaking tap | Aircon servicing | Power trip | Door lock | Wall mounting | Others | All posts

Leaking tap

woody needs help

help with my tap !



Posted by admin on 11/1/2026, 11:38:11 PM

Was this helpful? 👍 (1)

Edit post | Delete post

Posts

ikea

Search

Popular: Leaking tap | Aircon servicing | Power trip | Door lock | Wall mounting | Others | All posts

Others

Furniture assembly

stuck with building my new cupboard from ikea ...
any advice??



Posted by test-customer-1 on 11/4/2026, 9:14:27 PM

Was this helpful? 👍 (0)

Edit post | Delete post

DIY Community Forum

Viewing replies

Button to collapse/expand replies for each post

Leaking tap

woody needs help

help with my tap !



Posted by admin on 11/1/2026, 11:38:11 PM

Was this helpful? 👍 (1)

Replies (4) Show replies

Write a reply

Add reply



Posted by admin on 11/1/2026, 11:38:11 PM

Was this helpful? 👍 (1)

Replies (4) Hide replies

woody

By test-customer-1 on 11/2/2026, 5:17:30 PM

Edit reply Delete reply

r u sure woody needs help?

By admin on 11/2/2026, 5:31:02 PM

woody is too good to need help

By admin on 11/2/2026, 5:31:14 PM

Woody does not need help , if anyone can fix this , it is him

By WoodyDaBest on 11/2/2026, 9:41:42 PM

Write a reply

Add reply

DIY Community Forum

Updating a post or reply

Edit your post

Title

tap tricks

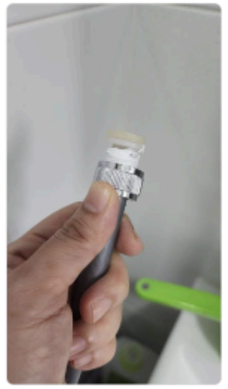
Category

Leaking tap | Aircon servicing | Power trip | Door lock | Wall mounting | Others

Details

Is this the right way to tape around a leaking tap???

These images are already attached and cannot be changed:



Save changes | Cancel

Aircon Leaking Water

Should I seek help for this or can I fix this manually by myself?



Posted by test-customer-1 on 11/2/2026, 9:38:12 PM

Was this helpful? 👍 (1)

Edit post | Delete post

Replies (1) Hide replies

i think you should seek help from one of the aircon services or else your room will be flooded :0

By WoodyDaBest on 11/2/2026, 9:40:29 PM

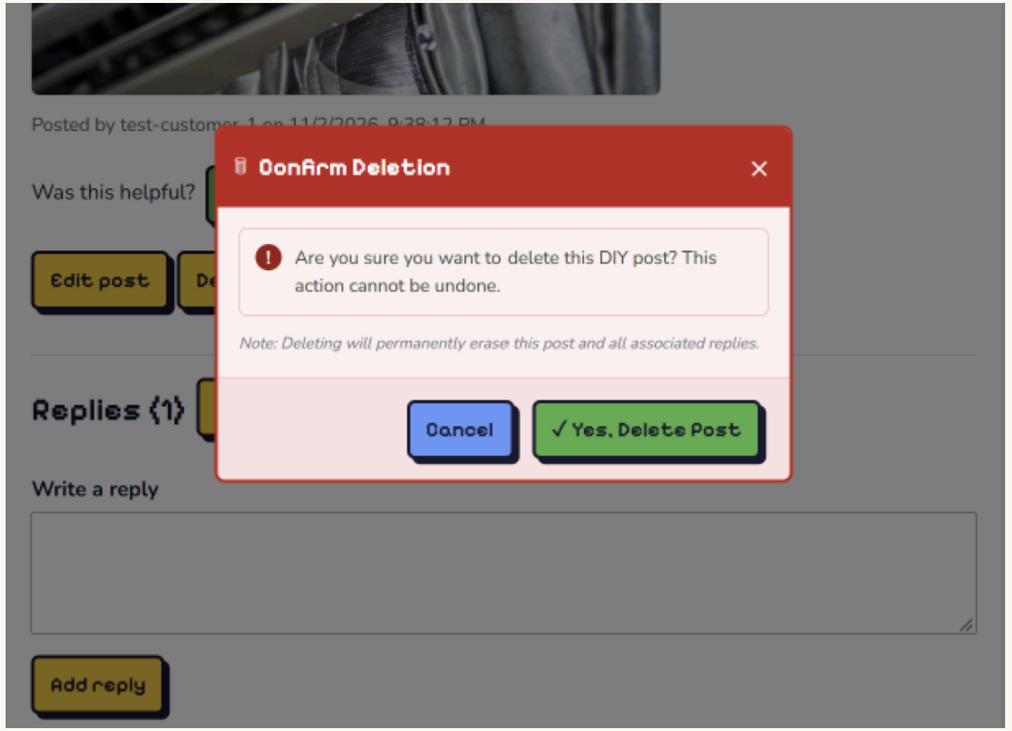
Edit reply

i think you should seek help from one of the aircon services or else your room will be flooded :0

Save reply | Cancel

DIY Community Forum

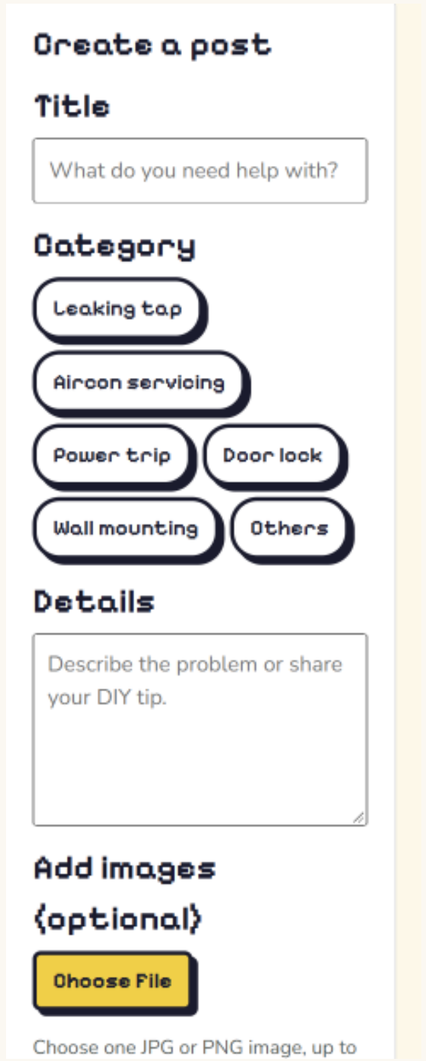
Deleting a post or reply

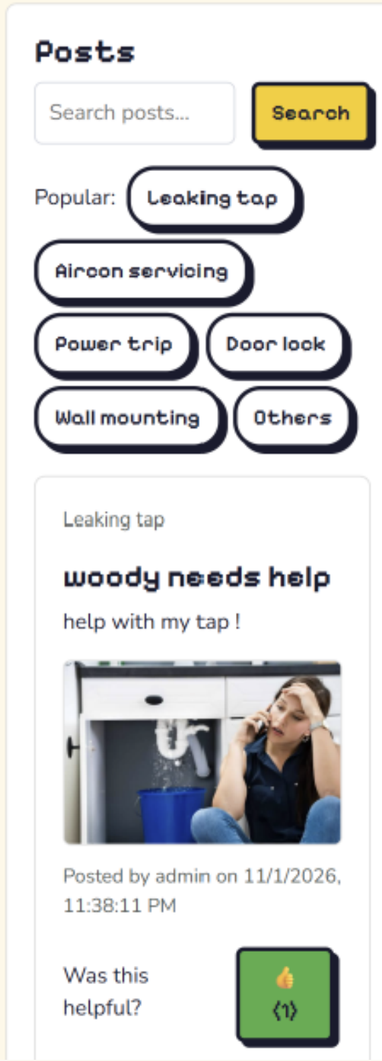


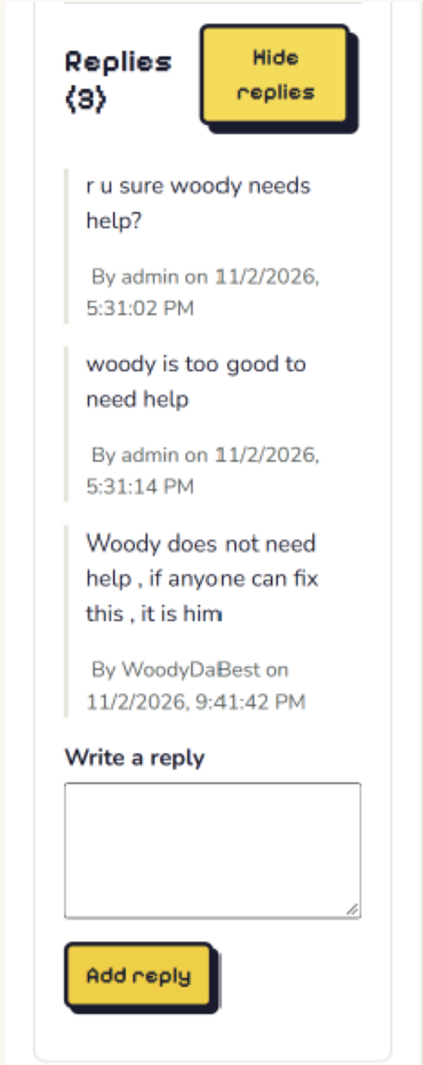


DIY Community Forum

Responsiveness on mobile







07

DIY Tool Shop

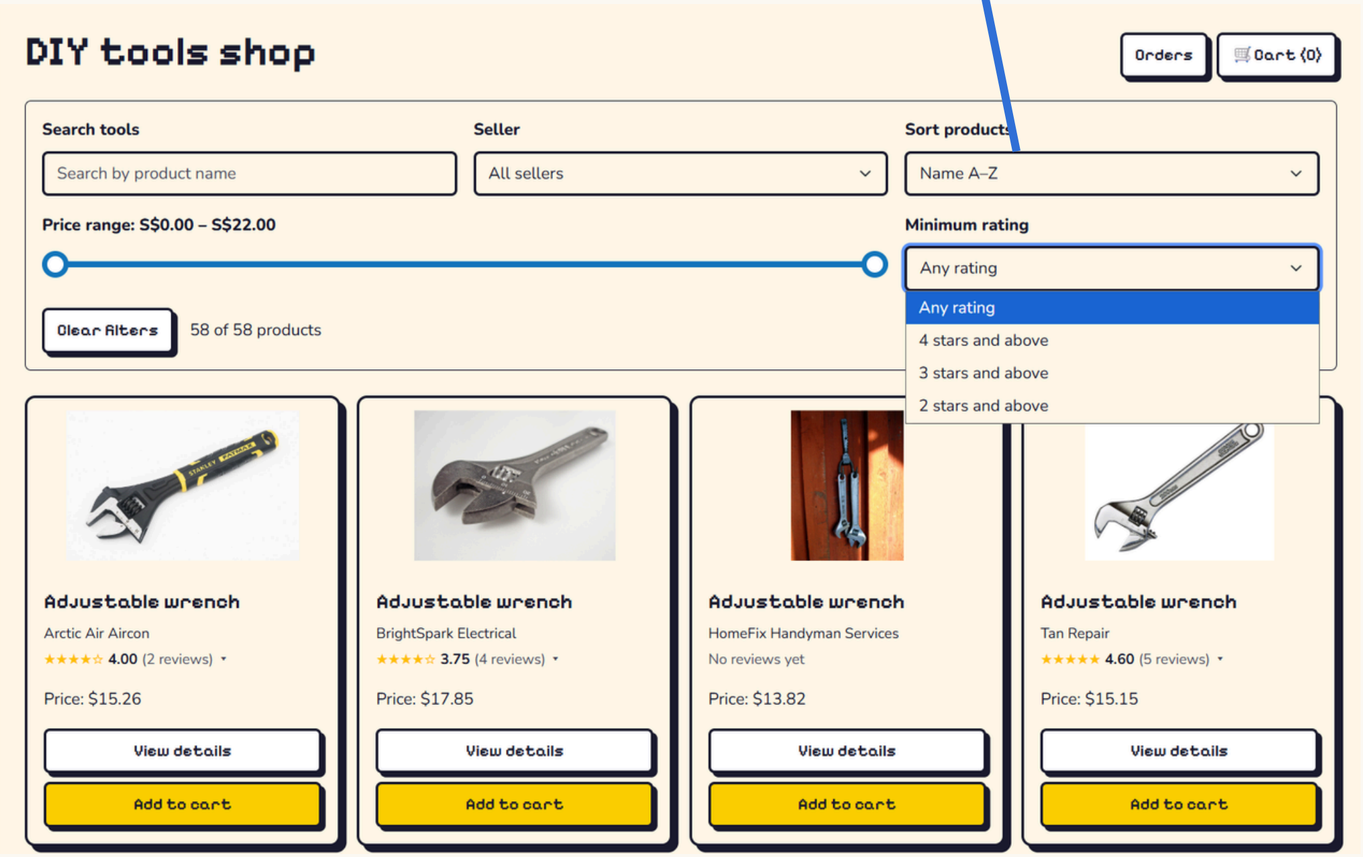
Shop

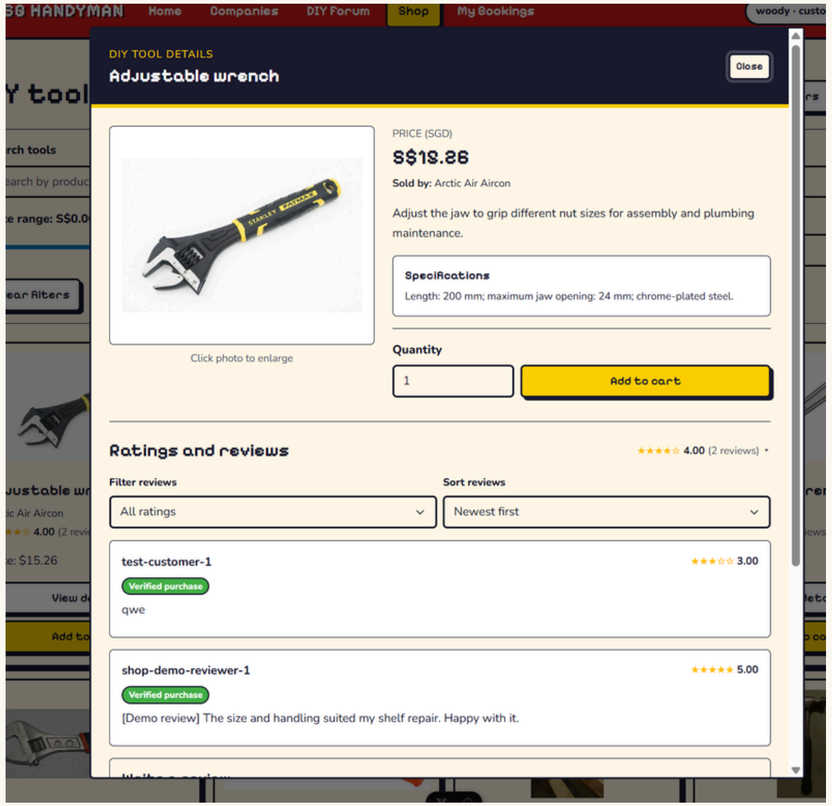
1. Searching/Navigating Product Page
2. Adding in to cart and checkout
3. View Orders progress and more
4. Giving reply and rating

Shop

Includes
- Name
- highest/lowest price
- ratings(high-low)

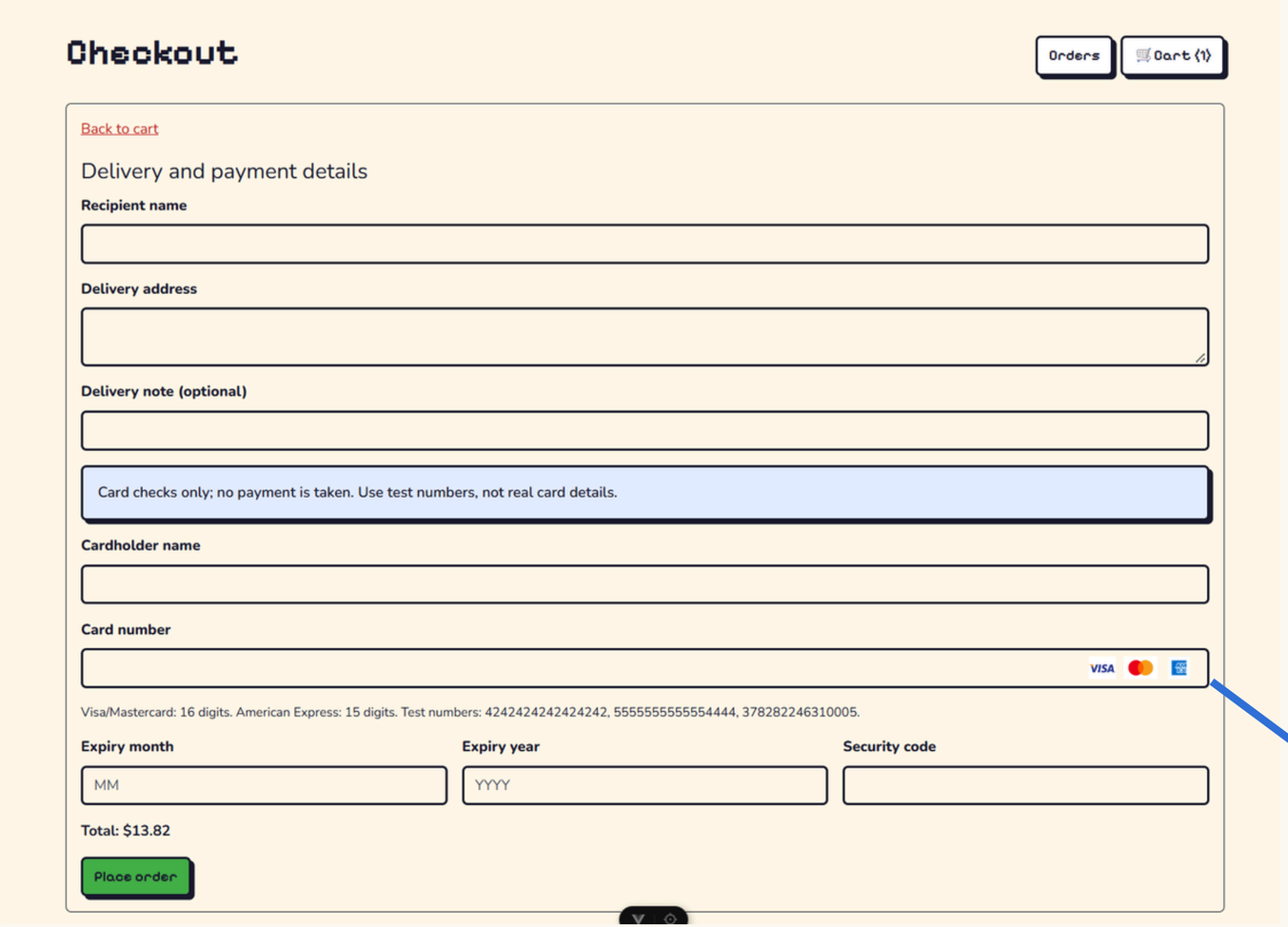
Searching/Navigating Product Page

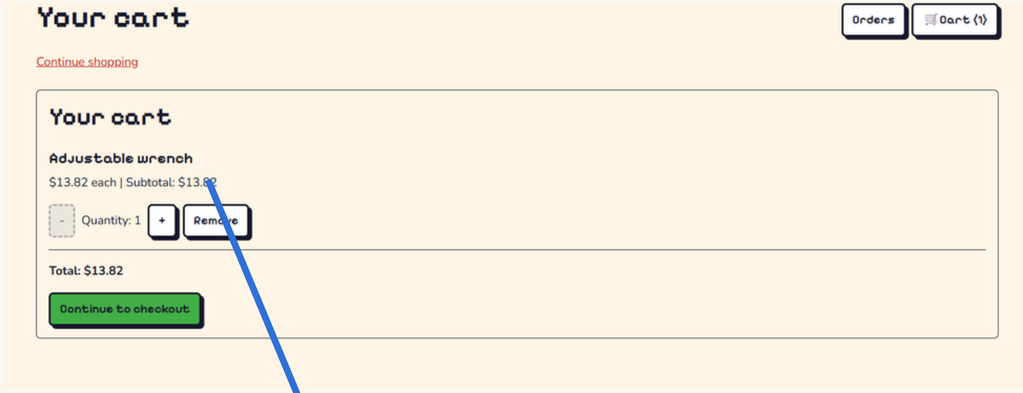




Shop

Adding in to cart and checkout



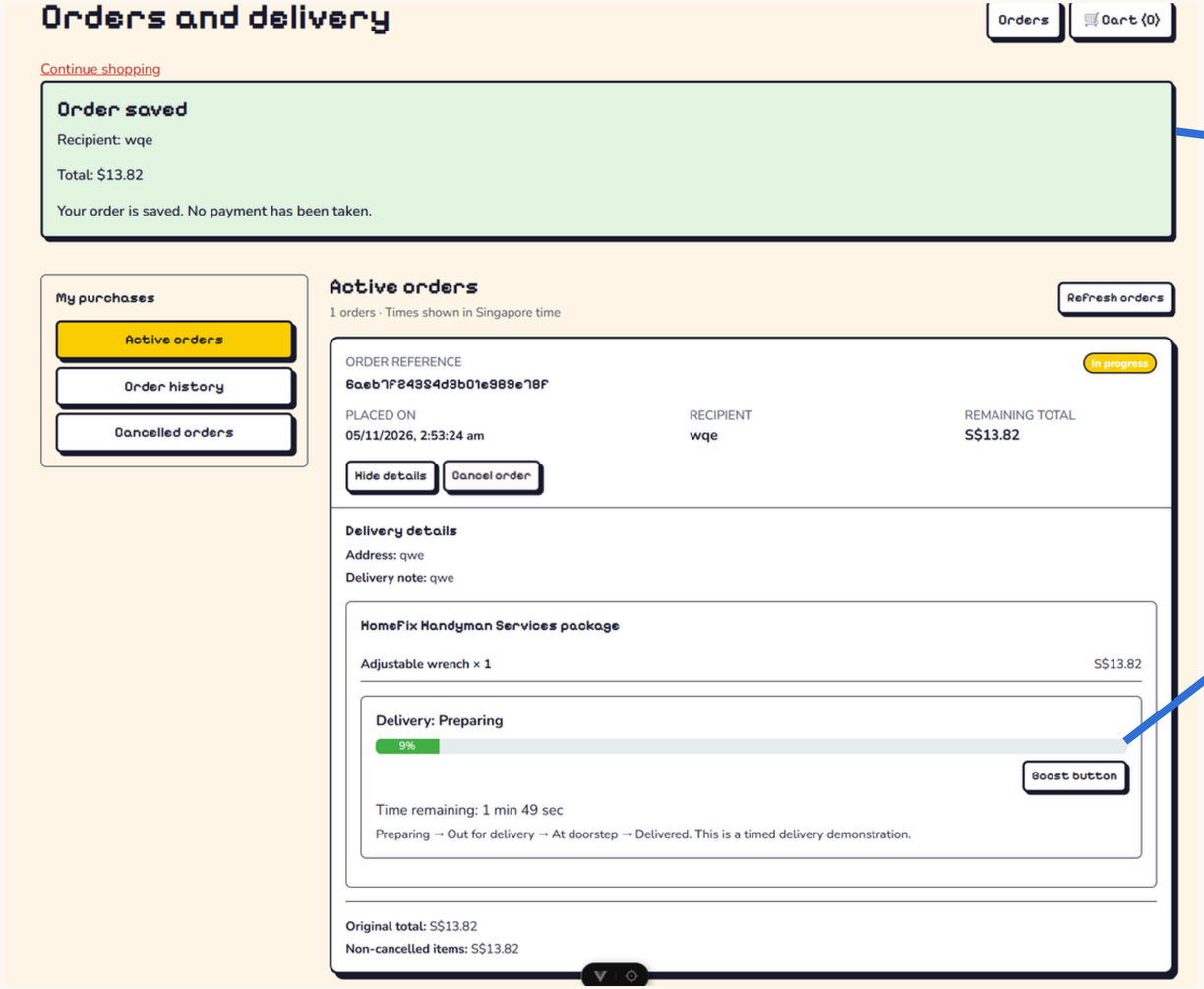


Direct changing to the product amount in the cart

On time reactive to the card number key in to change card icons

05 Shop

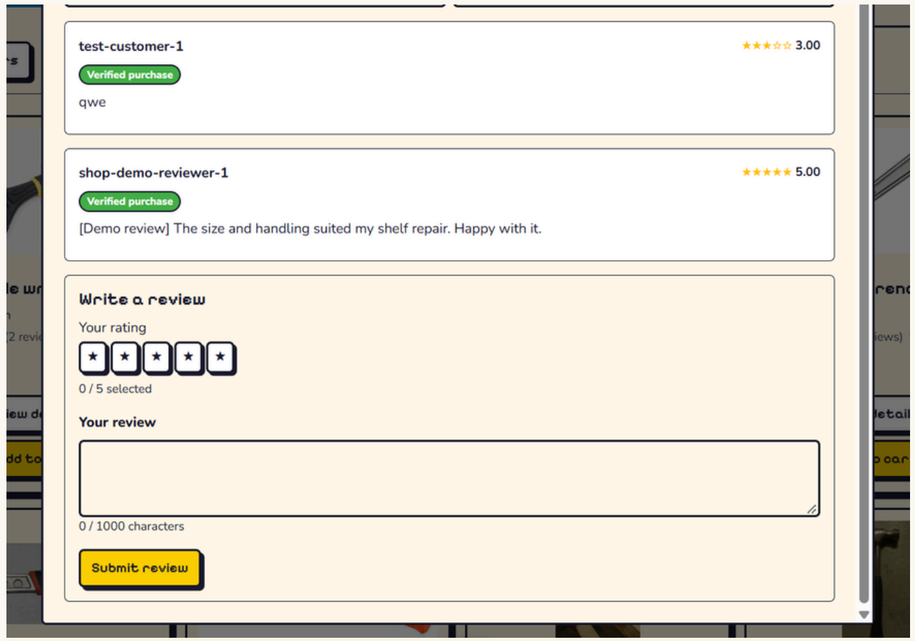
View Orders progress and more



Receipt After a action on most recent order related action
EG cancel/add order

Real time delivery Progress bar

Shop





Giving reply and rating

Each user is permitted to review once per product

User is only allowed to review after purchasing the product

08
Conclusion

IS216 G5 · Team 8

Thank you!

Questions?



04
Testing

04 TESTING

Testing

00 *Add a label*	00 *Add a label*	00 *Add a label*

User journeys covered

List the end-to-end flows you tested

05
Beyond the Requirements

Beyond the requirements

1 Extra

2 Extra

3 Extra

4 Extra

06
Live demo

07
Challenges & Lessons